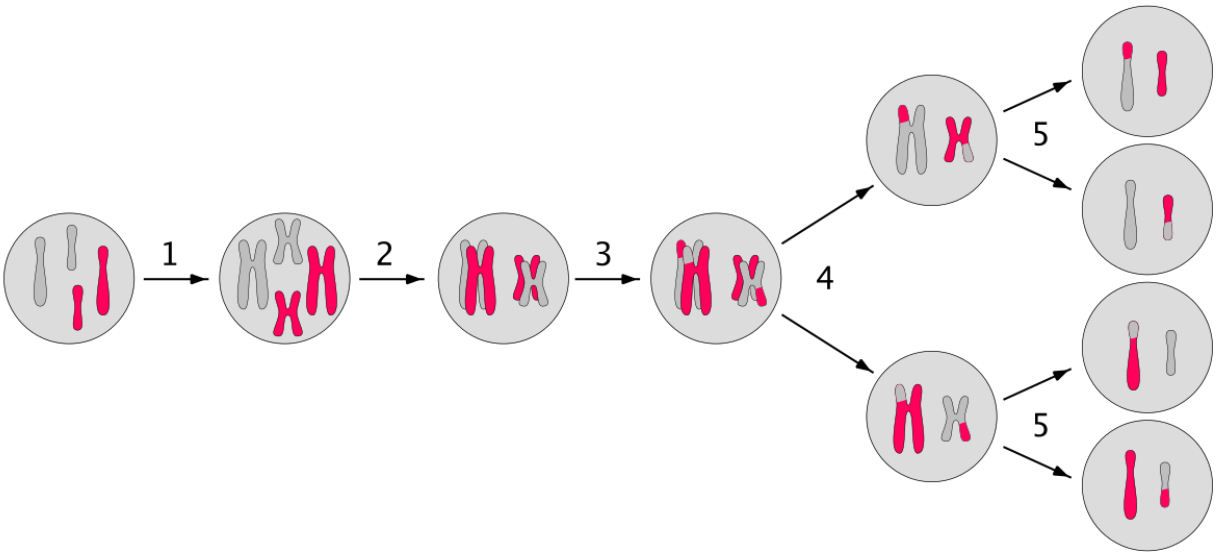
Science Olympiad CSE 2025

Designer Genes C KEY



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Directions:

- The test is broken into six parts, all of different lengths.
- Sections based on divisions in the rule book for this event.
- All MCQ are worth 1 point, True/False worth 2, FRQ and Multi-Select vary
- The test should be given in a 50 minute period.
- Total points: ____/150

Section 1 - Mendelian Genetics (21 pts)

1. Which of the following describes the Law of Independent Assortment best?
    a. Each gamete has an equal probability of obtaining either member of the gene
    b. Alleles for separate traits are passed independently of one another [+1]
    c. Genes being on the same chromosome means traits get passed separately
    d. Traits are passed independently of the phenotype of the parents

2. What decade did Gregor Mendel develop his theories of inheritance after experimentation with pea plants?
    a. 1780s
    b. 1860s [+1]
    c. 1910s
    d. 1890s

3. If the cross rr x Rr, approximately how many offspring in a batch of 200 be heterozygous?
    a. 100 [+1]
    b. 150
    c. 0
    d. 50

4. Draw a dihybrid cross for RrYy x RrYy - make sure to write the genotypes for each outcome [+6]

	RY	Ry	rY	ry
RY	RRYY	RRYy	RrYY	RrYy
Ry	RRYy	RRyy	RrYy	Rryy
rY	RrYY	RrYy	rrYY	rrYy
ry	RrYy	Rryy	rrYy	rryy

[+2 for 4x4 box WITH correct letters on top, +4 for crosses being correct]

5. Two genes, A and B, are located on the same chromosome. Explain how linkage between these genes could violate Mendel's Law of Independent Assortment. [+3]
    Linked genes violate Mendel's Law because they do not assort independently(Violates Law of Independent Assortment) [+3]

6. If a male with an X-linked dominant disorder has children with an unaffected female, what percentage of their sons will have the disorder?
    a. 0% [+1]
    b. 25%
    c. 50%
    d. 100%

7. Which inheritance pattern would result in a trait appearing in all male descendants of a male individual?
    a. X-Linked Recessive
    b. Mitochondrial
    c. Y-Linked [+1]
    d. Autosomal Recessive

8. Which inheritance pattern would result in a trait being passed from mother to all children?
    a. X-Linked Recessive
    b. Mitochondrial [+1]
    c. Y-Linked
    d. Autosomal Recessive

9. In a pedigree, two unaffected parents have multiple affected children. What is the most likely mode of inheritance?
    a. X-Linked Recessive
    b. Mitochondrial
    c. Y-Linked
    d. Autosomal Recessive [+1]

10. Which of the following information is not necessary in analyzing a pedigree?
    a. Genotype data from DNA Sequencing [+1]
    b. Phenotypic data from relative
    c. Known inheritance patterns
    d. Sex and relationship of individuals

11. A plant with the genotype AaBb is testcrossed with a plant that is aabb. If the A and B genes are 10 map units apart, what percentage of the offspring are expected to have a recombinant phenotype?
    a. 5%
    b. 10% [+1]
    c. 15%
    d. 20%

12. A rare autosomal recessive disorder affects 1 in 100,000 individuals in a large, randomly mating population. A healthy couple has a child who is affected by the disease. What is the probability, in percentage form, that both parents are carriers of the disease allele? [+2]

100% [+2] not much to show for work, since it can be solved logically with no math

13. Given that both parents from the following situation are confirmed carriers of this autosomal recessive condition, what is the probability their next offspring will be affected?
    a. 25% [+1]
    b. 50%
    c. 75%
    d. 100%

Section 2 - Mitosis and Meiosis (15 pts)

1. During which phase of mitosis do sister chromatids separate?
    a. Prophase
    b. Metaphase
    c. Anaphase [+1]
    d. Telophase

2. Which structure is primarily responsible for aligning chromosomes at the metaphase plate?
    a. Spindle fibers [+1]
    b. Centrosmere
    c. Kinetochore
    d. Centriole

3. If a diploid organism has a chromosome number of 2n = 6, how many chromosomes are in each daughter cell after meiosis?
    a. 2
    b. 3 [+1]
    c. 4
    d. 6

4. What phase does crossing over occur in meiosis?
    a. Prophase I [+1]
    b. Anaphase 1
    c. Metaphase II
    d. Metaphase I

5. Explain the difference between nondisjunction in meiosis I vs meiosis II. [+6]
    [Meiosis I nondisjunction → homologs fail to separate; all gametes are abnormal. +3
    Meiosis II → sister chromatids fail to separate; half gametes are abnormal.+3, +6 total]

6. The process on the front of this test is mitosis.

a. True
b. False [+2]

7. Which of the following correctly matches a given structure with its function or role only during meiosis I?
   a. Centrioles, form spindle poles in mitosis and meiosis
   b. Synaptonemal complex, helps pairing and recombination between homologous chromosomes [+1]
   c. Kinetochore, pulls apart sister chromatids in anaphase 1
   d. Microtubules, responsible for only cytokinesis in meiosis 1

8. In a diploid organism of 2n = 6, how many genetically unique gametes can be produced assuming only independent assortment (no crossing over)?
   a. 8 [+1]
   b. 6
   c. 64
   d. 12

9. During meiosis, homologous chromosomes fail to undergo synapsis in prophase 1. Which of the following is the most likely consequence of this failure?
   a. Reduced recombination frequency and normal gamete formation
   b. Increased recombination between sister chromatids
   c. Increased aneuploidy in the resulting gametes [+1]
   d. Premature separation of sister chromatids in anaphase I

Section 3 - Population and Evolutionary Genetics (37 pts)

1. List all the requirements for Hardy-Weinberg Equilibrium to be present in a population: [+5]
   i. No mutations
   ii. No immigration/migration
   iii. No natural selection
   iv. No sexual selection
   v. A large population

   [+5(+1 for each)]

2. Genetic drift is impossible in a Hardy-Weinberg world/
   a. True [+2]

b. False

3. Traits can't be lost from a population as long as there is genetic drift.
   a. True
   b. False [+2]

4. Write and label the two Hardy-Weinberg equations: [+4]
   $p^2 + 2pq + q^2 = 1$ (Genotype frequencies)
   $p + q = 1$ (Allele Frequencies)
   [+1 for each correct equation, +1 for each correct label, +4 total]

5. What is relative fitness? [+2]
   How many offspring an individual contributes [+1], compared to others of the population/species [+1]

6. In a population at Hardy-Weinberg equilibrium, the frequency of allele A is 0.7. What is the frequency of heterozygotes? Show your work. [+3]
   2pq[+1]
   2(0.7)(0.3) = 0.42
   [+1 for calculation and +1 for correct answer, +3 total]

7. A founder effect is most likely to occur when:
   a. Increase in territory for a species
   b. A few individuals go to a new isolated area [+1]
   c. Mutation rates increase
   d. Individuals mate with those close to them

8. The bottleneck effect leads to:
   a. Increased heterozygosity
   b. Reduced genetic diversity [+1]
   c. Greater allele frequency stability
   d. Increased gene flow

9. If 16% of a population is homozygous recessive (aa), what is the frequency of the dominant allele (A) under Hardy-Weinberg? [+4]
   $q^2 = 0.16$ [+1]
   $q = 0.4$ [+1]
   $p = 1 - 0.4 = 0.6$ [+1]
   Answer: 0.4 [+1] [total +4]

10. Which of the following would most likely result in parallel evolution due to duplicated genes?
    a. Paralogs [+1]
    b. Orthologs
    c. Homologs
    d. Natural Selection

11. Define and contrast genetic drift and gene flow and how they affect allele frequencies: [+4]
    Genetic drift: Random changes, esp. in small populations; reduces genetic variation. [+2]
    Gene flow: Migration of alleles between populations; increases similarity between populations. [+2]

12. Given the following aligned sequences, identify which is the outgroup and which two are the closest. [+4]
    i. Species A: ATCG
    ii. Species B: ATCA
    iii. Species C: AGCA
    iv. Species D: TCGA

    A and B are the closest [+2]
    D is the outgroup [+2]

13. Which of the following is most likely to reduce genetic variation within a population over time?
    a. Stabilizing selection
    b. Gene flow
    c. Migration
    d. Directional selection [+1]

14. In a population, a gene has two alleles, M and m. Individuals with the heterozygous genotype (Mm) have significantly higher fitness than either MM or mm homozygotes. What type of selection is this? [+2]
    Heterozygote advantage [+2], give [+1] for balancing selection

15. A population is found to have a higher-than-expected number of homozygotes compared to Hardy-Weinberg predictions. Which of the following is the most likely explanation?
    a. High mutation rate
    b. Mass migration
    c. Inbreeding [+1]
    d. Overdominance

Section 4 - Molecular Biology Of DNA (35 pts)

1. Explain why DNA strands are said to be antiparallel and describe the importance of 5' to 3' directionality during DNA replication. [+4]

   DNA strands are antiparallel because one strand runs 5' to 3' and the complementary strand runs 3' to 5'. [+2]
   DNA polymerase can only add nucleotides to the 3' end, so replication occurs in a 5' to 3' direction on each new strand. [+2]

2. Differentiate euchromatin from heterochromatin in terms of structure and function. Then give an example of each. [+4]

   Euchromatin: loosely packed, transcriptionally active, accessible to RNA polymerase (ex. housekeeping genes) [+1 for definition, +1 for example]

   Heterochromatin: tightly packed, transcriptionally inactive or silenced (ex. centromeric or telomeric DNA) [+1 for definition, +1 for example]

3. Which enzyme removes RNA primers and replaces them with DNA in prokaryotes?
   a. Ligase
   b. DNA polymerase 1 [+1]
   c. DNA polymerase III
   d. Primase

4. Which repair pathway is most likely used to correct UV-induced thymine dimers?
   a. Nucleotide Excision Repair [+1]
   b. Base Excision repair
   c. Mismatch Repair
   d. Homologous Recombination

5. Explain the difference between a frameshift mutation and a missense mutation. [+4]

   Frameshift: caused by insertions/deletions not in multiples of 3, alters the entire downstream reading frame. [+2]
   Missense: point mutation that changes one amino acid [+2]

6. Which of the above mutations is typically more harmful and why? [+2]

   Frameshift is usually more harmful because it affects all codons after the mutation [+2]

7. Which enzyme seals nicks by okazaki fragments?
   a. DNA Ligase [+1]
   b. Primase
   c. Okazise
   d. Gyrase

8. You observe a DNA fragment that is 30% thymine. What percent is cytosine? Show your work. [+2]

   30% T -> 30% A (A=T binding)
   Remaining 40% is C+G so 20% is C [+1 for answer +1 for work]

9. Why are plasmids useful in genetic engineering? [+4]

   Plasmids replicate independently, so they can carry foreign genes. This means they can be used to transform bacteria and express certain genes and proteins for engineering purposes. [+1 for mentioning independently, +3 for rest. +4 total]

10. During DNA replication, which enzyme relieves supercoiling ahead of the replication fork?
    a. Helicase
    b. DNA ligase
    c. Topoisomerase [+1]
    d. DNA polymerase III

11. Which DNA repair mechanism involves removal of a short stretch of the DNA strand followed by resynthesis using the complementary strand?
    a. Mismatch Repair [+1]
    b. Photoreactivation
    c. Double-strand break repair
    d. Base excision repair

12. What type of mutation changes a codon to a stop codon? [+2]

    Nonsense [+2]

13. Which of the following are processes that require DNA to be unwound or separated? (Select all that apply)
    a. Transcription [+1]
    b. DNA Replication [+1]
    c. Translation
    d. Protein folding
    e. DNA Repair [+1]
    f. RNA Splicing

14. What bonds connect the phosphate and sugar in DNA's backbone? [+2]

    Phosphodiester bonds [+2]

15. Which of the following are associated with heterochromatin? (Select all that apply)
    a. High transcriptional activity
    b. Gene splicing [+1]
    c. Acetylated histones
    d. Dense packing [+1]
    e. Telomeric regions [+1]

f. Active enhancers

Section 5 - Prokaryote Genetics (25 pts)

1. What is the correct order of events in bacterial transcription?
    a. Elongation, Initiation, Termination
    b. Initiation, Termination, Elongation
    c. Initiation, Elongation, Termination [+1]
    d. Termination, Elongation, Initiation

2. Which of the following describes what happens to the lac operon when lactose is present and glucose is low?
    a. Lac repressor is bound to the operator
    b. cAMP levels are low
    c. CAP binds to the promoter [+1]
    d. The operon is fully repressed

3. The trp operon is an example of:
    a. A positive inducible system
    b. A negative inducible system
    c. A positive repressible system
    d. A negative repressible system [+1]

4. Which of the following is not a function of sigma factors in prokaryotes?
    a. Initiate transcription
    b. Recognize promoter regions
    c. Catalyze peptide bond formation [+1]
    d. Assist RNA polymerase binding

5. Which directly blocks RNA polymerase in the lac operon when lactose is absent?
    a. CAP
    b. Promoter
    c. Operator
    d. Repressor protein [+1]

6. In prokaryotic translation, what component binds the Shine-Dalgarno sequence?
    a. Large ribosomal subunit
    b. tRNA
    c. RNA polymerase
    d. Small ribosomal subunit [+1]

7. A mutation causes the lac operon repressor in E.coli to lose its ability to bind to allolactose. Predict the effect of this mutation in the presence of lactose. [+2]
   The lac operon will remain repressed even when lactose is present. [+2]

8. Continuing from the above question, explain the molecular mechanism behind your prediction. [+4]
   Allolactase normally binds the repressor, causing it to change shape and fall off the operator. If the repressor can't bind allolactase, it will stay bound to the operator, preventing transcription, regardless of lactose presence. [+4 for full detailed explanation]

9. RNA polymerase synthesizes mRNA in the 3' → 5' direction on the DNA strand:
   a. True
   b. False [+2]

10. What terminates transcription in Rho-independent termination?
    a. Ribosome release
    b. Hairpin loop formation [+1]
    c. Rho helicase binding
    d. Exonuclease degradation

11. Where does the tRNA molecule's anticodon bind to?
    The mRNA codon [+2]

12. Some mRNAs can fold into structures that block the Shine-Dalgarno sequence.
    a. True [+2]
    b. False

13. What allows transcription and translation to occur simultaneously in prokaryotes? [+2]
    Lack of a nuclear membrane to separate the processes. [+2]

14. Which event occurs after transcription elongation begins?
    a. Binding of sigma
    b. DNA strands rewind behind polymerase [+1]
    c. Formation of pre-initiation structure
    d. Binding of Rho at promoter

15. State the Central Dogma of molecular biology. State one process that violates this dogma. [+3]
    Dogma: DNA → RNA → Protein [+1]
    Process Example: Reverse transcription seen in retroviruses [+2]

Section 6 - Technology and Techniques (17 pts)

1. Which of the following components is not required in a standard PCR reaction?
    a. Taq polymerase
    b. Primers
    c. DNA Ligase [+1]
    d. dNTPs

2. During the annealing step of PCR the temperature is typically?
    a. ~95°C
    b. ~60°C [+1]
    c. ~80°C
    d. ~35°C

3. What is the purpose of using dideoxynucleotides in Sanger sequencing? [+3]
    To terminate the synthesis of a growing DNA strand, resulting in partially replicated DNA fragments. [+3]

4. What is a limitation of Nanopore sequencing? [+3]
    Higher error rate due to base misinterpretation, short read lengths, GC content bias, etc [+3]

5. Give a reason why CRISPR is more widely used than RNAi in terms of gene editing? [+3]
    CRISPR edits DNA directly, heritable and precise edits, RNAi is only transient, etc [+3]

6. What distinguishes Illumina from Sanger sequencing?
    a. Illumina uses radioactive labeling, Sanger uses fluorescence
    b. Illumina sequences DNA by synthesis in clusters, Sanger sequences individual fragments [+1]
    c. Illumina relies on chain-terminating ddNTPs, Sanger does not
    d. Illumina can only sequence RNA, Sanger sequences DNA

7. What enzyme is responsible for extension in PCR, and what does it do? [+2]
    Taq polymerase [+1], synthesizes the DNA [+1]

8. What is the purpose of bridge amplification in Illumina sequencing?
    a. Terminate strand synthesis
    b. Amplify DNA fragments on a solid surface [+1]
    c. Translate RNA into cDNA
    d. Detect specific protein to DNA interactions

9. In RNA sequencing, mRNA is commonly converted into:

a. Proteins for analysis of expression
b. DNA via reverse transcriptase [+1]
c. Single stranded RNA problem
d. Fluorescently labeled proteins to map expression

10. Which molecule is used to silence a gene at the post-transcriptional level in RNAi?
a. siRNA [+1]
b. gRNA
c. ddNTP
d. sgRNA